TABLE I

BENCHMARK RESULTS GROUPED BY TASK DIFFICULTY AND FLOOR SCOPE. THE CLEAN SCHEDULE CONTAINS 270 SINGLE-FLOOR AND 38 CROSS-FLOOR SINGLE-TARGET EPISODES, AND 364 SINGLE-FLOOR AND 182 CROSS-FLOOR MULTI-TARGET EPISODES. SR AND SPL ARE PERCENTAGES; CT IS AVERAGED OVER SUCCESSFUL EPISODES. A DASH INDICATES NO SUCCESSFUL EPISODE. ONEMAP USES ITS ORDERED-CATEGORY PROJECTION. BOLD DENOTES THE BEST RESULT IN EACH TASK GROUP.

Task difficulty	Method	Single-floor			Cross-floor		
		SR↑ (%)	SPL↑ (%)	CT↓ (s)	SR↑ (%)	SPL↑ (%)	CT↓ (s)
Single-target	SpatialNav	8.15	5.89	22.35	–	–	–
	SpatialGPT	10.37	2.69	115.39	–	–	–
	MapGPT	27.78	14.35	42.95	18.42	13.08	81.57
	OneMap	52.59	35.45	77.32	21.05	20.30	49.88
	LogicFly	**97.78**	**87.28**	**8.19**	**94.74**	**86.03**	**8.99**
Multi-target	MapGPT	1.10	0.73	96.02	–	–	–
	OneMap	12.36	8.16	150.13	6.59	6.59	90.59
	LogicFly	**96.70**	**82.49**	**10.33**	**96.70**	**81.45**	**14.70**

and real-world experiments. All language-model components use Qwen3.8-Flash. All perception, planning, and recovery parameters are fixed across scenes and episode repetitions.

Runtime protocol. Each evaluation process is subject to a fixed time limit of 240 s. A timeout or execution error is counted as a failed episode. For LogicFly, CT is measured using its native completion timestamp and excludes one-time map preparation. For the comparison methods, CT is measured from process startup to task completion. CT is averaged over successful episodes only.

B. Comparison with Baseline Methods in Simulation

We compare LogicFly with HM3D adaptations of OneMap [22], MapGPT [6], SpatialGPT [7], and SpatialNav [28]. All methods run on the same 854-episode clean schedule. OneMap retains its native ordered-category projection, which omits relations, conditional branches, verified absence, recovery, and return constraints. All baselines and LogicFly are evaluated using the same frozen strict full-task evaluator. SpatialGPT and SpatialNav do not successfully complete any cross-floor episode, resulting in an SR of 0%; therefore, their remaining cross-floor metrics are not reported.

As shown in Table I, LogicFly consistently outperforms the compared methods across target count and floor scope. On single-target tasks, it reaches 97.78% SR, 87.28% SPL, and 8.19 s CT on single-floor episodes, and 94.74% SR, 86.03% SPL, and 8.99 s CT on cross-floor episodes. On the harder multi-target tasks, it maintains a 96.70% SR on both floor scopes, with SPL values of 82.49% and 81.45% and CT values of 10.33 s and 14.70 s for single-floor and cross-floor episodes, respectively. SpatialGPT and SpatialNav have no successful multi-target episodes and are therefore omitted from that block. Across the full 854-episode schedule, LogicFly completes 828 episodes for a 96.96% strict full-task SR and an overall SPL of 0.8394.

Figure 8 presents a qualitative execution example in a three-floor Habitat environment. LogicFly completes the cross-floor route and all required subtasks, with terminal views providing visual evidence for each successful outcome.

All baselines retain their native planning, action, and stopping interfaces. OneMap receives the protocol-defined ordered-category projection, whereas MapGPT, SpatialGPT, and SpatialNav receive the frozen task specification through their native interfaces. Timeouts and algorithm failures remain unsuccessful terminal outcomes.

C. Ablation Studies

TABLE II

ABLATION RESULTS ON THE MULTI-TARGET SUBSET OF 546 EPISODES. SR AND SPL ARE PERCENTAGES, AND CT IS AVERAGED OVER SUCCESSFUL EPISODES.

Variant	SR↑ (%)	SPL↑ (%)	CT↓ (s)
w/o CPTG	64.84	61.85	35.63
CPTG + fixed views	**96.70**	78.03	11.65
CPTG + JTVP	**96.70**	**82.15**	**6.39**

We evaluate CPTG and JTVP on the 546 multi-target episodes from the frozen six-scene HM3D schedule. This subset includes ordered, freely schedulable, conditional-branch, verified-absence, semantic-recovery, spatial-relation, cross-floor, and target-disambiguation tasks. Running all three variants yields 1638 end-to-end multi-target trials.

All variants are executed end to end using the native Habitat runner, online open-vocabulary verification, and the same strict evaluator. They share identical scene representations, start poses, motion constraints, termination rules, and evaluation settings. *w/o CPTG* removes task dependencies, conditional rules, and task-conditioned candidate generation, retaining only target labels, coarse room/floor scopes, generic observation constraints, and fixed language order. *CPTG + fixed views* retains CPTG and task-order optimization but uses one independently selected fixed observation pose per task. *CPTG + JTVP* jointly optimizes task ordering and observation viewpoints. The evaluator requires every required outcome and effective order, forbids visits to skipped branches, and checks spatial/reference constraints.

As shown in Table II, removing CPTG reduces multi-target SR to 64.84% and SPL to 61.85%, while increasing successful-task CT to 35.63 s. Without an explicit task graph, the system cannot reliably enforce precedence relations, activate observation-dependent branches, or distinguish required